SOLAR ORACLE WALKMAN — INDEPENDENT ENGINEERING PROTOTYPE

POWER

檔案： power.kicad_sch

CONTROL

檔案： control.kicad_sch

MEASUREMENT

檔案： measurement.kicad_sch

LOADS

檔案： loads.kicad_sch

ALWAYS_ON

檔案： always_on.kicad_sch

SWITCHING

檔案： switching.kicad_sch

K1 RELAY VARIANT: see README.md for contacts and coil timing.
Glass66x66mm; optical opening50x50mm (routed corners); ENIG contacts5x50mm.
Three PHB1.5F/5V packs in parallel; no DAC / Teensy / audio adaptor / LLM.
Set JP1 OFF BEFORE USB; LOLIN USB VBUS and its5V header are the same node.
DSSC Voc-0.65V / Isc-34mA are separate endpoints; Pmpp unmeasured. CAD consistency is not bench qualification.

LOLIN S3 MINI sockets / 16 MOSFET loads / AEM10941 / Adafruit 6366

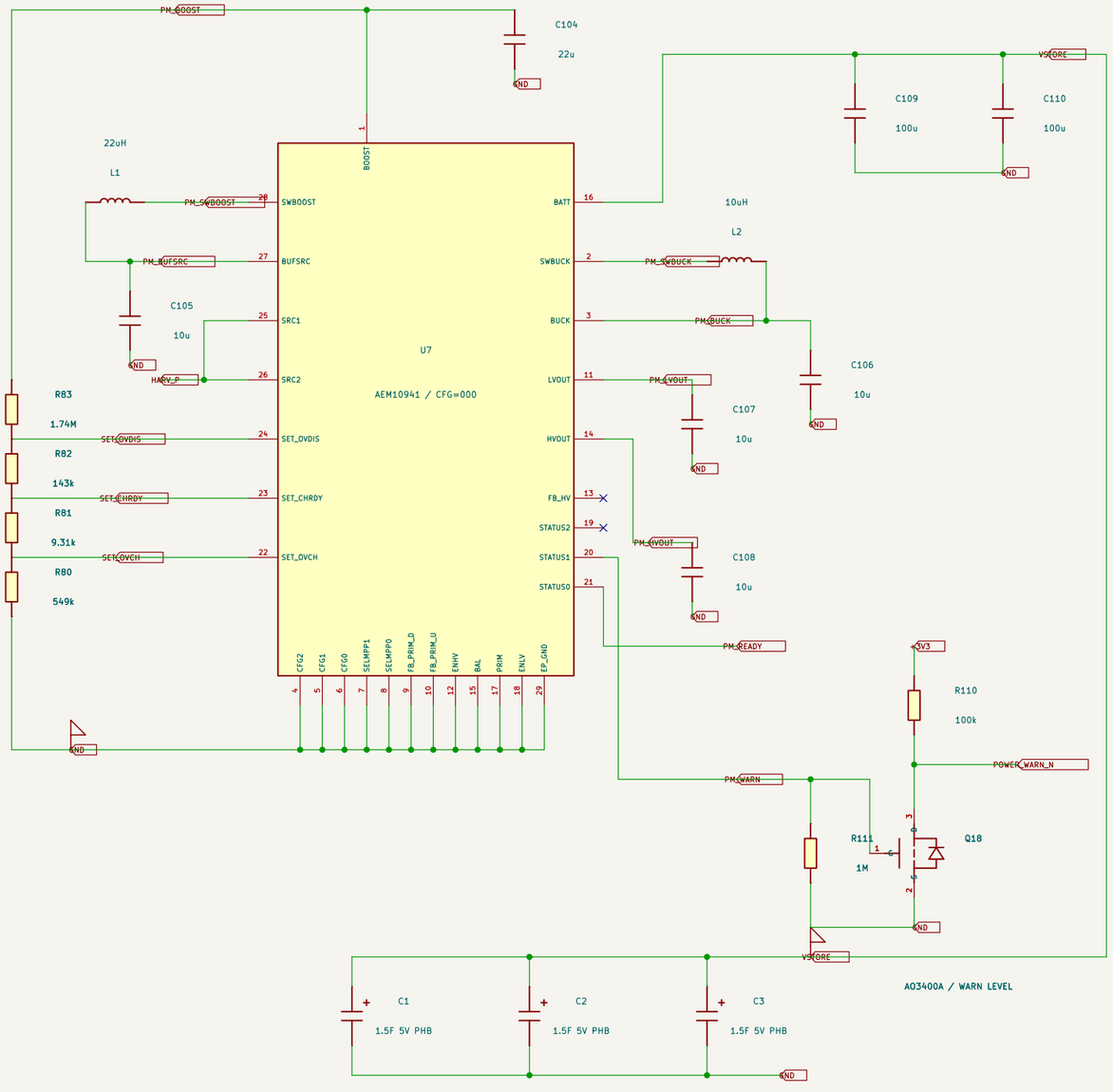
Wise Mouse Culture

Sheet: /
File: solar-oracle-walkman-relay.kicad_sch

Title: Solar Oracle Walkman — K1 Relay

Size: A3	Date:	**Rev: K1-RELAY**
KiCad E.D.A. 10.0.5		Id: 1/7

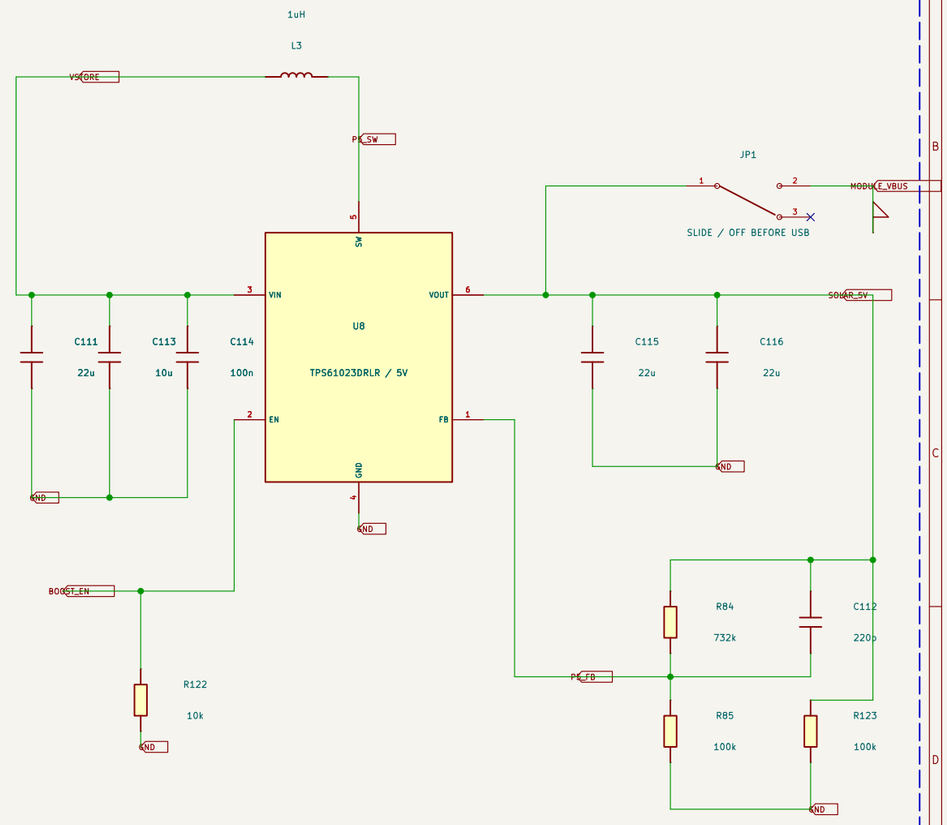
1. AEM10941 — HARVEST / CHARGER



C1/C2/C3: three complete PHB packs in PARALLEL: 4.5F / 5V.

CFG=000: MPPT=70% Voc. Bottom configuration pins wired individually to GND.

2. TPS61023 — CONTROLLED 5V



R84/R85/C112: feedback group. R122: EN bias; R123: output bleed.

JP1 is still the removable header: OPEN BEFORE CONNECTING USB.

Solar Oracle Walkman

Sheet: /POWER/
File: power.kicad_sch

Title: POWER: AEM harvest + TPS61023 5V

Size: A3	Date:	**Rev: G20260913**
KiCad E.D.A. 10.0.5		Id: 2/7

CONTROL A — LOLIN S3 / BUTTON / MAIN I2C

CONTROL B — TWO 74HC595 / FIXED CHANNEL ORDER

CONTROL C — ADAFRUIT 6366 / SPI / CONTROL

Cross-sheet interfaces keep their original net names. GPIO map GPIO20260920; approved J2 routability permutation; channel order unchanged.

QA is pin15; QB..QH are pins1..7. QH' pin9 is the cascade output. Zero-hot / one-hot only.

J3 pin2 stays NC. SD select (pin10) held HIGH. RESET / BUSY / SRAM_CS retained.

Solar Oracle Walkman	
Sheet: /CONTROL/	
File: control.kicad_sch	
Title: CONTROL: LOLIN / 595 cascade / ePaper	
Size: A3	Date:
KiCad E.D.A. 10.0.5	Rev: CM20260913
	Id: 3/7

MEAS A — DUT / DIVIDER / TMUX1511 ISOLATION

J1 DSSC ZEBRA 5x50 — DSSC+ → DUT_P; DSSC− → GND

R87 1M (VSTORE → S1)

R88 330k (VCAP_DIV → S2)

U6 TMUX1511PWR / OFF ISOLATION

- VDD ← +3V3; C128 100n to GND
- S1 ← VSTORE; D1 → SENSE_SW
- S2 ← VCAP_DIV; D2 → VCAP_ADC
- D3 ← I2C_SDA; S3 → ADC_SDA
- D4 ← I2C_SCL; S4 → ADC_SCL
- SEL1 ← SENSE_EN; SEL2 ← SENSE_EN; SEL3 ← SENSE_EN; SEL4 ← SENSE_EN
- GND → GND

R98 10k (SENSE_EN → GND)

U6 isolates DUT sense, capacitor divider AND ADC I2C. Analog supply must be valid before SENSE_EN.

MEAS B — ADC / DIFFERENTIAL FILTERS / ADC I2C

U3 ADS122C04IPWR / 0x40

- AVDD ← ANALOG_3V3; C124 100n to GND
- DVDD ← ANALOG_3V3; C125 100n to GND
- C129 10u (ANALOG_3V3 to GND)
- C120 10n (GND to SENSE_SW)
- R94 1k (SENSE_SW → AIN0); C118 100n (AIN0 to AIN1)
- R95 1k (GND → AIN1)
- R96 1k (LOUT → AIN2); C119 100n (AIN2 to AIN3)
- R97 1k (BIAS_1V25 → AIN3)
- AIN0 ← ADC_V_P; AIN1 ← ADC_V_N; AIN2 ← ADC_I_P; AIN3 ← ADC_I_N
- SDA ↔ ADC_SDA; SCL ↔ ADC_SCL
- R112 4.7k, R113 4.7k (SDA, SCL → ANALOG_3V3)
- A0, A1, DGND, AVSS → GND
- REFN → GND
- REFP ← REF_2V5; C121 100n (REF_2V5 to GND)
- RESET_N ← ADC_RESET_N; R99 10k (ANALOG_3V3 to ADC_RESET_N)
- DRDY_N → ADC_DRDY_N; R114 10k (ADC_DRDY_N to ANALOG_3V3)

PGA bypass / gain1 / external 2.5V reference; sequential V/I conversions, not simultaneous.

MEAS C — SHUNT / KELVIN / INA190

U4 INA190A4IDCKR / G=200

- VS ← ANALOG_3V3; C127 100n to GND
- R17 0.100 1% CAL (SHUNT_H → GND)
- R92 100 (SHUNT_H → ISENSE_P); R93 100 (ISENSE_N → GND)
- C117 22n (ISENSE_P to ISENSE_N)
- IN+ ← ISENSE_P; IN− ← ISENSE_N
- OUT → VCAP_ADC
- REF → GND
- GND → GND

Take both sense branches at R17 lands; keep return current out of the R93 sense spur.

MEAS D — REF2025 / BIAS

U5 REF2025AIDDCR

- VIN ← ANALOG_3V3; C126 100n to GND
- EN → GND
- VREF → C122 1u to GND; also → BIAS_1V25
- VBIAS → C123 1u to GND
- GND → GND

MEAS E — ANALOG POWER

U9 TPS22919DCKR / ANALOG POWER

- IN ← +3V3; C130 1u to GND
- ON ← ANALOG_EN; R115 10k (ANALOG_EN to GND)
- OUT → ANALOG_3V3; C131 1u to GND
- QOD → GND
- NC → GND
- GND → GND

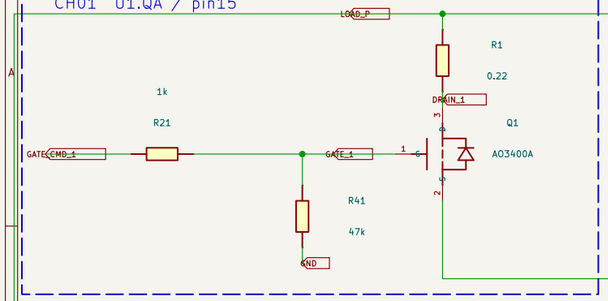
Solar Oracle Walkman

Sheet: /MEASUREMENT/
File: measurement.kicad_sch

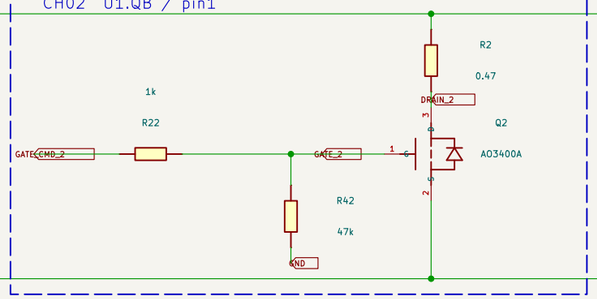
Title: MEASUREMENT: isolation / ADC / INA / reference / supply

Size: A3	Date:	Rev: CM20260913
KiCad E.D.A. 10.0.5		Id: 4/7

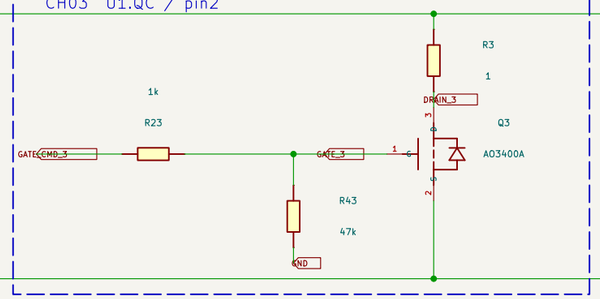
CH01 U1.QA / pin15



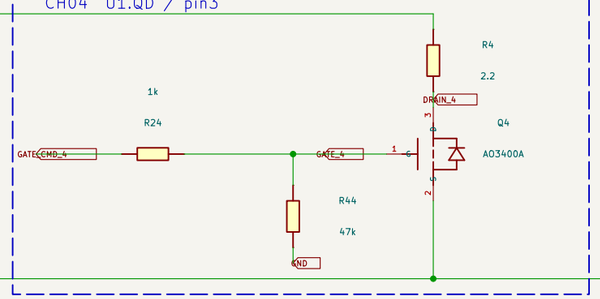
CH02 U1.QB / pin1



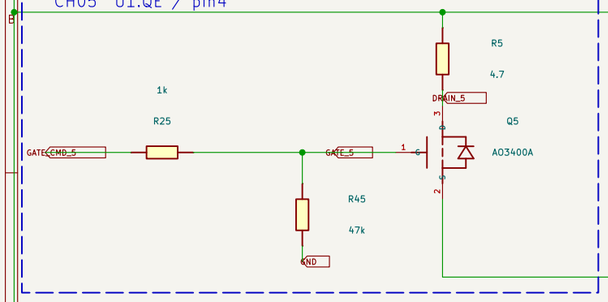
CH03 U1.QC / pin2



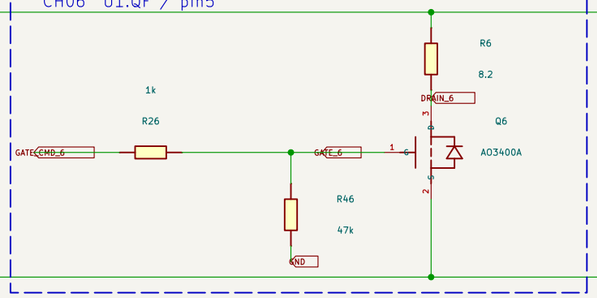
CH04 U1.QD / pin3



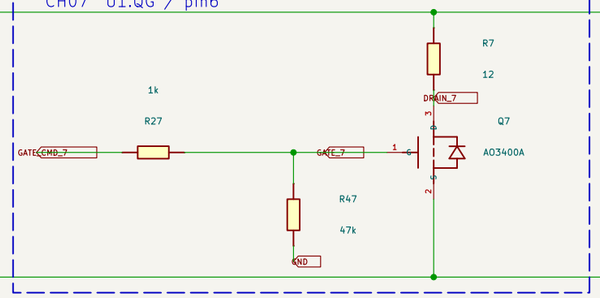
CH05 U1.QE / pin4



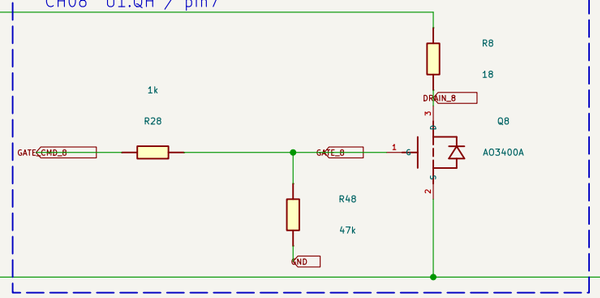
CH06 U1.QF / pin5



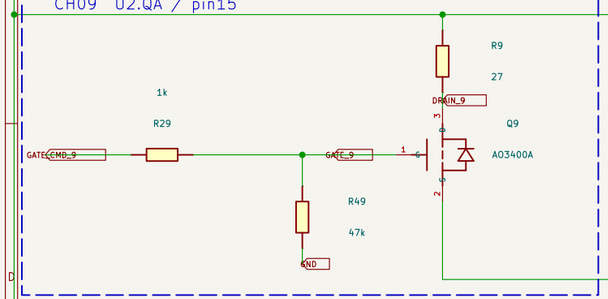
CH07 U1.QG / pin6



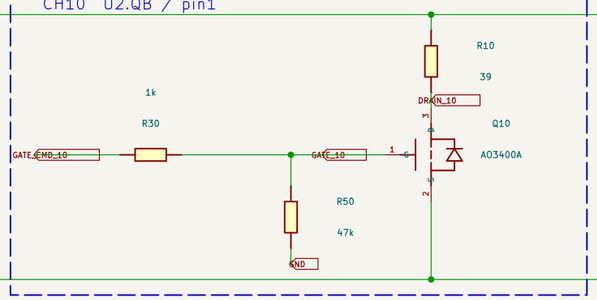
CH08 U1.QH / pin7



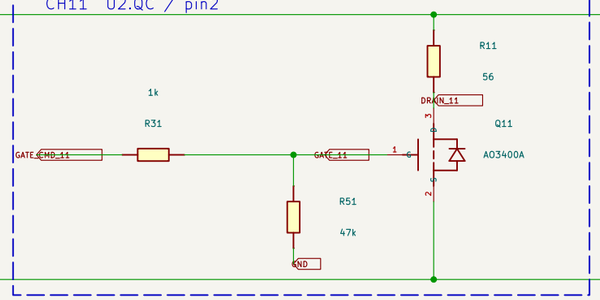
CH09 U2.QA / pin15



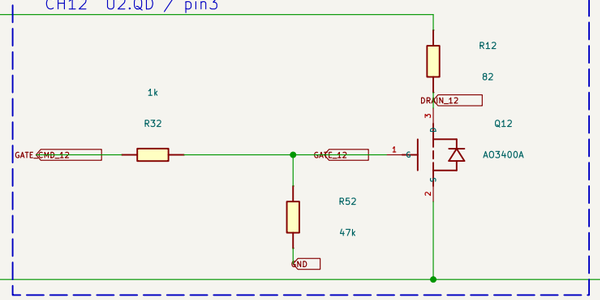
CH10 U2.QB / pin1



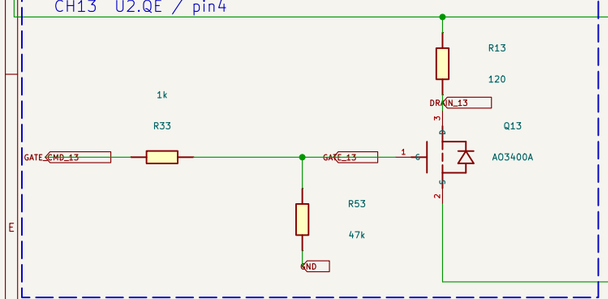
CH11 U2.QC / pin2



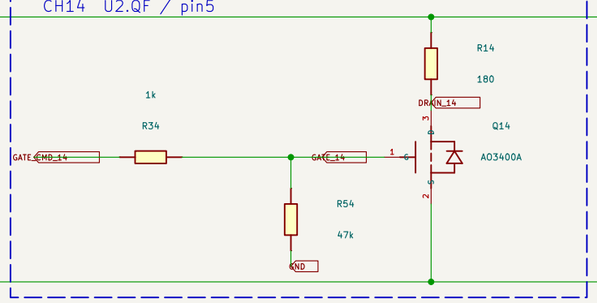
CH12 U2.QD / pin3



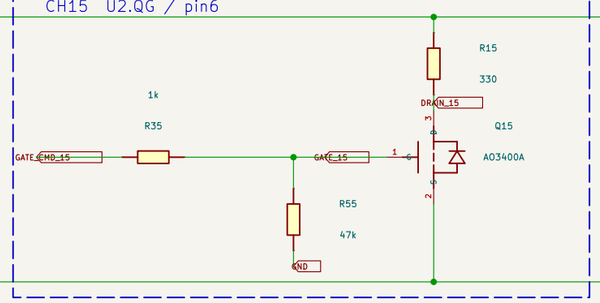
CH13 U2.QE / pin4



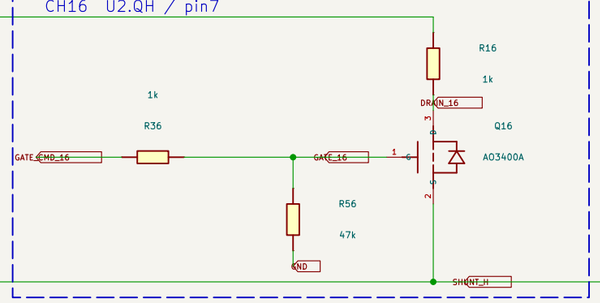
CH14 U2.QF / pin5



CH15 U2.QG / pin6



CH16 U2.QH / pin7



One channel = load resistor + AO3400A + 1k gate resistor + 47k pull-down. Select a dashed tile to collect its four symbols.

MOS sources -> SHUNT_H -> R17 on MEASUREMENT. Gate pull-downs -> GND below R17; do not merge the two return nets.

Latch all-zero before enabling OE. Zero-hot or one-hot only; no PWM. Lowest-resistance points are near-short, not exact Isc.

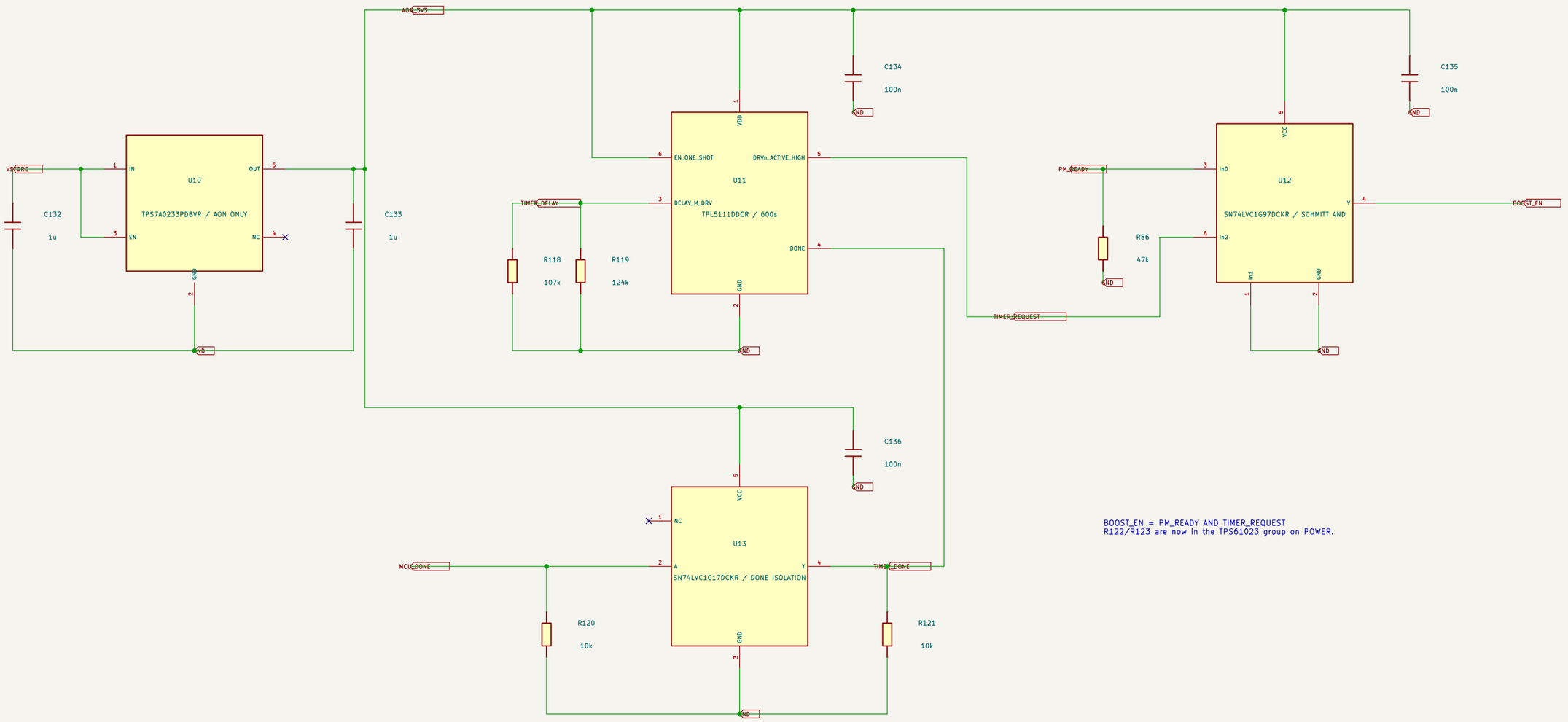
Solar Oracle Walkman

Sheet: /LOADS/
File: loads.kicad_sch

Title: LOADS: 16 fixed resistor channels / explicit wiring

Size: A3	Date:	**Rev: L20260913**
KiCad E.D.A. 10.0.5		Id: 5/7

3. ALWAYS_ON — SUPPLY / TIMER / READY / DONE



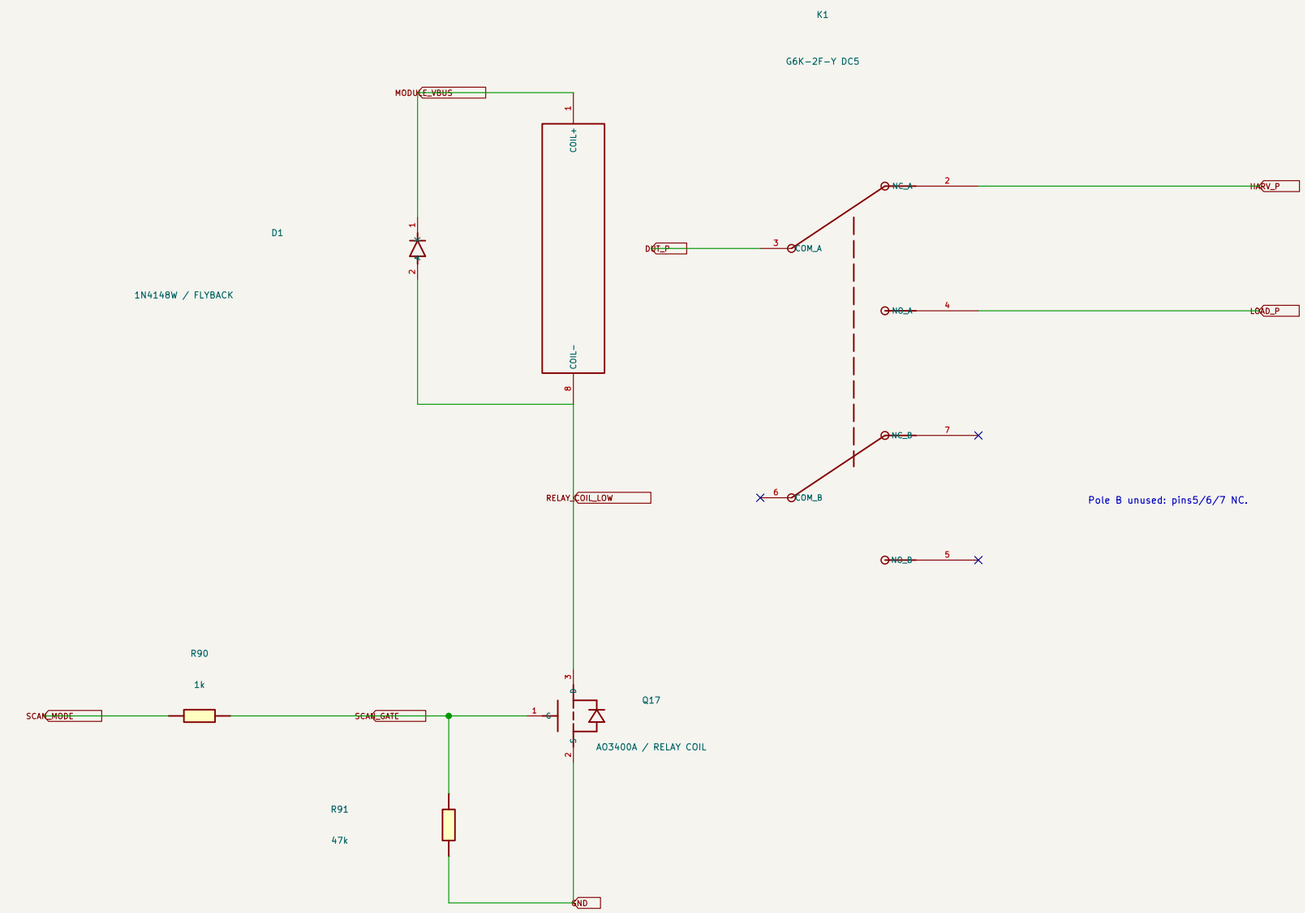
BOOST_EN = PM_READY AND TIMER_REQUEST
R122/R123 are now in the TPS61023 group on POWER.

U10 + C132/C133: AON supply. U11 + C134/R118/R119: timer.

U12 + C135/R86: READY permission. U13 + C136/R120/R121: DONE interface.

Solar Oracle Walkman		
Sheet: /ALWAYS_ON/		
File: always_on.kicad_sch		
Title: ALWAYS_ON: READY / timer / DONE		
Size: A3	Date:	**Rev: G20260913**
KiCad E.D.A. 10.0.5		Id: 6/7

K1 MECHANICAL RELAY — DEFAULT HARVEST / ENERGIZED MEASUREMENT



SCAN_MODE LOW / no power: COM3 -> NC2 -> HARV_P; LOAD_P disconnected.

SCAN_MODE HIGH: COM3 -> NO4 -> LOAD_P; HARV_P disconnected.

Coil: 5V / 237ohm / nominal21.1mA. Pin1 positive, pin8 low-side driver.

Disable 595 outputs and latch zero before switching. Wait for contact settling before sampling.

Flyback diode extends release time. Measure settling/release; 3ms relay spec is not a diode-clamped bound.

Solar Oracle Walkman		
Sheet: /SWITCHING/		
File: switching.kicad_sch		
Title: SWITCHING: K1 relay / coil driver / flyback		
Size: A3	Date:	**Rev: K1-RELAY**
KiCad E.D.A. 10.0.5		Id: 7/7